Dual Graph Learning Affinity Propagation for Multimodal Remote Sensing Image Clustering

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***Abstract*— Multimodal remote sensing image recognition aims to identify a category of land cover for every pixel with consistency and complementary information provided by different modalities. Most existing methods perform land cover recognition in a supervised manner with explicit label guidance. It is challenging to perform recognition without label guidance due to the complex spatial distribution and modality incompatibility, especially for large-scale data. In this article, we propose a dual graph learning affinity propagation (DGLAP) method for multimodal remote sensing image clustering. Based on the consistent spatial distribution from local regions, the proposed method learns an $N \times M$ consensus anchor graph from N denoised pixels and M anchors by adaptive weighting different modalities along with projection learning. Meanwhile, an optimal $M \times M$ compressed consensus anchor graph is learned from the updated anchors in different modalities with diverse adaptive contributions and connectivity constraint. Since $M \ll N$, clustering results can be efficiently obtained according to affinity propagation from the pseudolabeled anchors to the pixels without additional steps. An alternating optimization algorithm is devised to solve the proposed formulation. This is the first attempt to propose a ultraefficient graph-based clustering method with linear time complexity $\mathcal{O}(N)$ and low time cost for large-scale multimodal remote sensing data. Extensive experiments on three datasets demonstrate the superiority of the proposed method over the state-of-the-art methods in both efficacy and efficiency. The code is released at https://github.com/ZhangYongshan/DGLAP.**

***Index Terms*— Efficient graph-based clustering, local region guidance, multimodal remote sensing, projection learning.**

I. Introduction

With the development of remote sensing technology, a scene over the Earth surface can be recorded by different sensors or platforms to provide plentiful information at different perspectives for better Earth observation [1], [2], [3]. Given an observation area, hyperspectral (HS) images contain hundreds of spectral bands that are beneficial to land cover classification [4], [5]. Digital surface model (DSM) images consist of height information to describe the ground elevation of different objects [6]. Besides, there are other widely used remote sensing data, such as synthetic aperture radar (SAR) images providing structure information, multispectral (MS) images providing spatial information, and light detection and ranging (LiDAR) images providing altitude information [7]. Multimodal remote sensing data comprise at least two different modalities that contain shared information and provide complementary information for each other. For a complex urban scene, HS images show their limitation in identifying land covers with extremely similar spectral signatures. To enable more accurate recognition, DSM images with height information can be used as complementarity to identify the asphalt on the road or on the building.

Multimodal remote sensing image recognition tries to explore consistency and complementarity from different modalities to identify land cover for every pixel [8], [9], [10]. To achieve classification, shallow and deep models are two categories with diverse data fusion strategies. For shallow models, feature concatenation [11] is the most simple strategy to feed the concatenated multimodal features into a classifier. Besides, multimodal data are well-explored by graph fusion [12] and multiple kernel learning [13]. These shallow models perform classification based on the original data that are unable to extract discriminative features. For deep models, early fusion, middle fusion, late fusion, encoder–decoder fusion, and cross fusion are effective strategies to fuse multimodal data. To be specific, early fusion, middle fusion, and late fusion [6] make a decision about where to fuse data into the concatenated features. Encoder–decoder fusion [14] learns the shared features from all the modalities by multiple encoders and decoders, while cross fusion [6] extracts compact features across modalities with plug-and-play modules. Based on the developed techniques, effective deep models are in blossom. Advanced models use network stream [6], attention mechanism [15], and adversarial learning [7] for accurate classification. However, they perform classification relying on ground-truth information acquired by extensive expertise. Pixel annotation is labor-intensive and time-consuming for a complex urban scene. It is difficult to achieve recognition without label guidance.

Graph-based clustering is promising to group samples with similar characteristics by exploring their geometric structures

Manuscript received 9 January 2024; revised 3 May 2024; accepted 6 June 2024. Date of publication 19 June 2024; date of current version 5 July 2024. This work was supported in part by the National Natural Science Foundation of China under Grant 62106241, and in part by the Knowledge Innovation Program of Wuhan-Shuguang under Grant 2023010201020335. *(Corresponding author: Xinwei Jiang.)*

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Digital Object Identifier 10.1109/TGRS.2024.3416459

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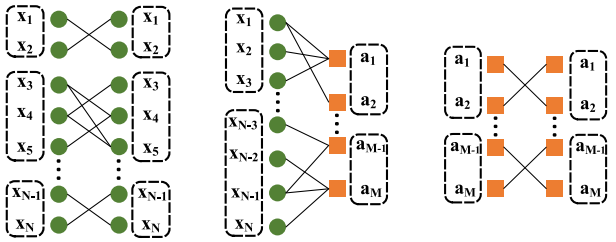


Fig. 1. Examples of different graphs. A solid circle denotes a sample and a solid square indicates an anchor. Given N samples and M anchors, $M \ll N$. (a) Ordinary graph with the size of $N \times N$. (b) Anchor graph with the size of $N \times M$. (c) Compressed anchor graph with the size of $M \times M$.

without using explicit labels [16], [17], [18]. There are a number of effective graph-based clustering methods. Traditional methods are developed on deep subspace clustering [19], broad learning [20], and nonlinear diffusion [21] to perform partition on an $N \times N$ ordinary graph describing the relationships among N pixels, as shown in Fig. 1(a). When the number of pixels is very large, they are confronted with awful computation and memory cost. Given M anchors ($M \ll N$), anchor graph reduces the graph size from $N \times N$ to $N \times M$ to deal with large-scale data, as shown in Fig. 1(b). Based on anchor graph, efficient methods achieve clustering on large-scale data with nonnegative relaxation [22], sequential procedure [23], and fuzzy theory [24], respectively. Although these methods obtain superior results, they are designed for single-modal data instead of multimodal data. Besides, they ignore spatial information for learning. Given multimodal data, low-rank constraint [25] and hypergraph regularization [26] have been explored in recent works. They require quadratic complexity that is inapplicable for large-scale data. To address this issue, superpixel segmentation technique is applied to rescale learning at superpixel level instead of pixel level [27]. However, clustering results may be greatly affected by the pixel-to-superpixel relationships since a superpixel and pixels within this superpixel are assumed to share the same label. To directly perform pixel-level clustering, two existing multimodal deep models have been recently proposed by designing multisensor networks [28] and multimodal fusion transformer [29]. Although these networks achieve superior results, their structure and training should be carefully designed and their interpretability is unclear. Thus, we still focus on graph-based clustering for multimodal remote sensing data.

Based on these observations, there is a challenging problem needed to solve in graph-based clustering for large-scale multimodal remote sensing data. That is, how to uncover the geometric structures from multimodal data with consistency and complementary information by exploring spatial information and discriminative representations with acceptable complexity and time cost. Feature enhancement can be achieved by spatial denoising and projection learning, while anchor graph is effective for large-scale problems. To directly obtain clustering results, rank constraint is usually imposed on a graph to learn exactly connected components. It is still time-consuming with $\mathcal{O}(M^2N + M^3)$ complexity to perform singular value decomposition (SVD) on an $N \times M$ anchor graph. How to further reduce the time complexity and consumption to learn the clustering structure from a graph? Instead of direct manipulating the $N \times M$ anchor graph, as shown in Fig. 1(c), we can learn a much small graph with exactly C connected components to explore the clustering structure.

Inspired by these considerations, in this article, we propose a DGLAP method for multimodal remote sensing image clustering. Fig. 2 illustrates the flowchart of the proposed method. To explore consistent spatial information, multimodal images are stacked into a larger tensor to generate superpixels reflecting homogeneous regions. According to the generated superpixels, denoising and anchor generation for different modalities are conducted with local spatial information. Based on N denoised pixels and M anchors from different modalities, an $N \times M$ consensus anchor graph is learned along with projection learning and adaptive modality weighting. Meanwhile, an optimal $M \times M$ compressed anchor graph with connectivity constraint is learned from projected anchors by adaptive weighting different modalities. The original and compressed anchor graphs are jointly learned in a mutual reinforcement manner. An alternating optimization algorithm is designed to solve the proposed formulation. A clustering inference strategy with affinity propagation is presented to propagate the clustering labels from anchors to pixels based on the maximum affinity recording in the original anchor graph without additional clustering steps. Experiments demonstrate the superiority of the proposed method in both efficacy and efficiency.

The main contributions of this article are listed as follows.

1) As far as we know, DGLAP is the first proposed efficient graph-based method for pixelwise multimodal remote sensing image clustering. It is promising and suitable for large-scale multimodal data.
2) We simultaneously learn an $N \times M$ anchor graph and an $M \times M$ compressed anchor graph with exactly connected components in a one-pass manner. It accelerates the optimal graph construction with a smaller graph.
3) We present an alternating optimization strategy to solve the proposed formulation. It exhibits high efficiency with linear time complexity of $\mathcal{O}(N)$.
4) We introduce a clustering inference strategy with affinity propagation to assign clustering labels for all the pixels based on the optimal compressed anchor graph with pseudolabeled anchors and the original anchor graph. It is highly efficient and effective.

The rest of this article is organized as follows. Section II reviews the related works. Section III presents the proposed DGLAP in detail. Comprehensive experiments are demonstrated in Section IV. Finally, Section V concludes the article.

II. Related Works

A. Multimodal Remote Sensing Image Classification

To deal with multimodal remote sensing data, shallow and deep models have been proposed by exploiting complementary and consistency information from different modalities for

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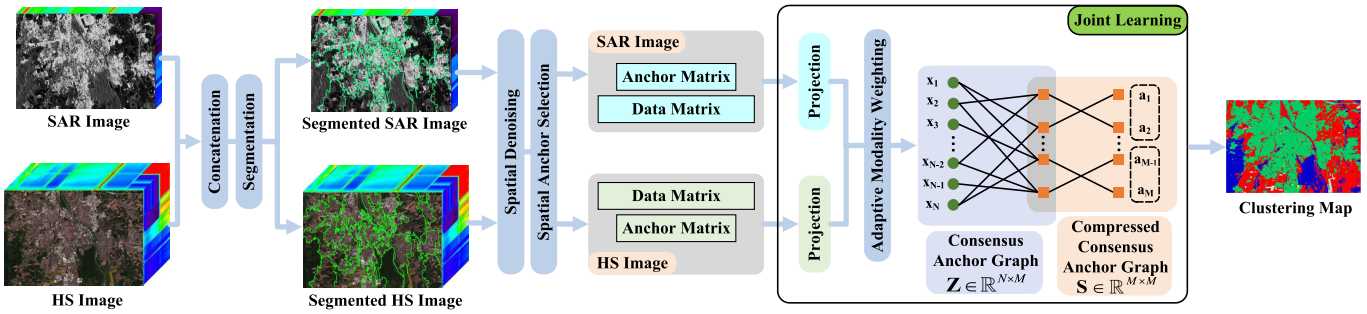


Fig. 2. Illustration of the proposed DGLAP. It performs segmentation to obtain the consistent superpixels for different modalities. Based on the segmented images, spatial denoising and anchor selection are conducted to jointly learn the original and compressed consensus anchor graphs along with projection learning and adaptive modality weighting. Here, we take two modalities as an example.

classification. For example, Liao et al. [12] proposed a graph-based fusion method to fuse HS and LiDAR data based on dimension reduction. For local climate zone classification, Yokoya et al. [11] simply stacked MS and OSM data together and fed them into a traditional classifier. Based on different data fusion strategies, Chen et al. [30] integrated multisource data for accurate land cover classification. Besides, other shallow models have been proposed by considering more advanced fusion strategies [13], [31]. Inspired by deep learning technique, Hong et al. [6] developed a multimodal deep learning framework to investigate different fusion strategies. To explore the complementarity, Gao et al. [7] presented an adversarial complementary network with four encoders in the generator and one shared decoder in the discriminator. Based on HS and LiDAR data, Li et al. [15] designed a fusion network to learn enhanced representations by spatial, spectral, and multiscale attention mechanisms. Deep models provide promising results because of their potential in discriminative feature extraction [32], [33].

Even though the existing classification methods achieve superior performance, they are inapplicable when ground-truth information is absent. Since accurate and sufficient labels are difficult to obtain, unsupervised methods are more flexible in real-world applications even when labels are not available. Thus, we try to develop an efficient and effective clustering method to analyze multimodal remote sensing images in an unsupervised manner.

B. Graph-Based Clustering for Remote Sensing Images

As an unsupervised learning scheme, graph-based clustering is effective to explore geometric structures for remote sensing data. Han et al. [34] designed a spectral–spatial network to learn discriminative representations for graph-based clustering. Based on the spatial distribution, Wang et al. [18] learned an anchor graph to achieve fast spectral clustering for HS data. To handle large-scale data, Zhao et al. [35] rescaled clustering at superpixel level and then performed pixel partition. These methods perform clustering on remote sensing data with only one data source. When data are described with multiple sources, they show the inefficacy of complementarity exploration. To make the best of spectral bands, Tian et al. [25] applied spectral partition to obtain different views and used morphological features as another view for clustering. With hypergraph regularization, Huang et al. [26] learned a shared clustering structure from original spectral features and extracted spatial features. Benefitting from the geometric structure, these methods provide superior results on small data. When data are in large volume, they are ineffective because of the required high computational consumption $\mathcal{O}(N^3)$ and huge memory cost $\mathcal{O}(N^2)$. To solve this issue, Chen et al. [27] rescaled learning at superpixel level and assigned labels for pixels to be the same as the corresponding superpixel. When a superpixel contains many pixels with diverse land covers, clustering results will be inferior and degraded.

Although the existing graph-based clustering methods obtain promising results, they are not very suitable for large multimodal remote sensing data because of their potential weaknesses analyzed above. To deal with large multimodal remote sensing data, we propose an ultraefficient and effective graph-based method to directly perform pixel-level clustering with consistency and complementary information as well as spatial information at very low time cost.

III. Proposed Method

A. Formulation

1) Spatial Denoising and Anchor Generation: Due to the wicked observation condition, multimodal remote sensing data collection may be noisy. To eliminate noise interference, we perform denoising with spatial information via superpixel segmentation technique. Considering the promising performance in both efficacy and efficiency, we adopt the entropy rate superpixel (ERS) segmentation method [36] to adaptively generate homogeneous regions based on the spatial distribution and texture complexity. Given a multimodal remote sensing dataset $\{\hat{\mathcal{X}}^v \in \mathbb{R}^{W\times H\times B_v}\}_{v=1}^{V}$ with V modalities and $W \times H$ pixels, each modality contains B_v channels to describe the recorded scene. To explore consistent spatial information within diverse local regions, we first stack different modalities into a larger tensor $\hat{\mathcal{X}} \in \mathbb{R}^{W\times H\times B}$ ($B = \sum_{v=1}^{V} B_v$) along with the third dimension. To capture the major information and achieve segmentation, we conduct principal component analysis (PCA) [35] on $\hat{\mathcal{X}}$ to obtain the first principal component I_f and perform ERS on I_f to generate M superpixels as

$$I_f = \cup_{i=1}^{M} \mathcal{H}_i, \quad \text{s.t. } \mathcal{H}_i \cap \mathcal{H}_j = \varnothing, \quad (i \neq j) \tag{1}$$

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where $\mathcal{H}_i$ is the ith superpixel. Since pixels at the same position in different modalities share the same ground truth, the above segmentation results can be aligned with multimodal images to reflect the whole segmented images.

Similar pixels with the same label are mostly distributed in one local region [37]. Under the generated superpixel guidance, multimodal remote sensing data denoising is able to reduce noisy information and smooth homogeneous regions with spatial information. Specifically, for the ith pixel in the vth modality $\hat{\mathbf{x}}_i^v$, we choose its L nearest neighbors with minimum Euclidean distance from the same superpixel denoted by $\Omega(\hat{\mathbf{x}}_i^v) = \{\mathbf{y}_1^v, \mathbf{y}_2^v, \ldots, \mathbf{y}_L^v\}$ and perform denoising to obtain the denoised pixels $\mathbf{x}_i^v$ as follows:

$$\mathbf{x}_i^v = \frac{1}{L+1}\left(\hat{\mathbf{x}}_i^v + \sum_{j=1}^{L} \mathbf{y}_j^v\right). \tag{2}$$

After denoising with (2), multimodal pixels are spatially distributed in a good consistency. The denoised multimodal data are reshaped into matrices $\{\mathbf{X}^v = \{\mathbf{x}_1^v, \mathbf{x}_2^v, \ldots, \mathbf{x}_N^v\} \in \mathbb{R}^{B_v \times N}\}_{v=1}^V$ $(N = W \times H)$ for subsequent learning.

Anchor selection strategy is vital to determine representative anchors to cover all samples for anchor-graph-based methods. Traditional random and k-means strategies [22] suffer from randomness problem and computational burden, respectively. Besides, they are unable to generate high-quality anchors for remote sensing images without considering the special data characteristics. In view of this, we choose the jth initial anchor $\mathbf{a}_j^v$ for the vth modality by averaging all the denoised pixels from the jth superpixel as

$$\mathbf{a}_j^v = \frac{1}{N_j}\sum_{i=1}^{N_j} \mathbf{x}_i^v \tag{3}$$

where N_j denotes the number of pixels and $\mathbf{x}_i^v$ indicates the ith denoised pixel for the jth superpixel in the vth modality. Since there are M superpixels, anchor matrices for all the modalities contain M anchors denoted by $\{\mathbf{A}^v = \{\mathbf{a}_1^v, \mathbf{a}_2^v, \ldots, \mathbf{a}_M^v\} \in \mathbb{R}^{B_v \times M}\}_{v=1}^V$. Compared with random and k-means strategies, (3) is promising to select representative anchors with spatial information for remote sensing data.

2) Anchor Graph Projection Learning: Compared with an $N \times N$ ordinary graph, an $N \times M$ anchor graph can reduce computational complexity and memory consumption from $\mathcal{O}(N^2)$ to $\mathcal{O}(NM)$ [18]. Since $M \ll N$, anchor graph shows its potential and efficiency to address large-scale problems. Based on multimodal data $\{\mathbf{X}^v\}_{v=1}^V$ and multimodal anchors $\{\mathbf{A}^v\}_{v=1}^V$, we hope to construct a consensus anchor graph $\mathbf{Z} \in \mathbb{R}^{N \times M}$ from different modalities to record their shared pixel-to-anchor neighboring relationships in an adaptive weighting manner. Since multimodal remote sensing data always consist of a high-dimensional HS image and other low-dimensional images [6], we also hope to perform projection learning to alleviate unbalanced dimensions of different modalities and improve their representation capability. Considering these situations, anchor graph projection learning constructs $\mathbf{Z}$ by adaptively assigning pixels to neighboring anchors and fusing different modalities in the projected subspace as

$$\begin{aligned}\min_{\mathbf{W}^v, \mathbf{U}^v, \mathbf{Z}, \alpha_v} & \sum_{v=1}^{V} \frac{1}{\alpha_v} \sum_{i=1}^{N}\sum_{j=1}^{M} \left\|\mathbf{W}^{vT}\mathbf{x}_i^v - \mathbf{u}_j^v\right\|_2^2 z_{ij} + \delta\|\mathbf{Z}\|_F^2 \\ \text{s.t.} \quad & \mathbf{W}^{vT}\mathbf{S}_t^v\mathbf{W}^v = \mathbf{I}, \quad \mathbf{z}_i^T\mathbf{1} = 1,\ z_{ij} \geq 0,\ \boldsymbol{\alpha}^{\mathrm{T}}\mathbf{1} = 1 \\ & \alpha_v > 0 \end{aligned} \tag{4}$$

where $\mathbf{W}^v \in \mathbb{R}^{B_v \times R_v}$ is the projection matrix for the vth modality to reduce the dimension from B_v to R_v, $\mathbf{U}^v = \mathbf{W}^{v\mathrm{T}}\mathbf{A}^v \in \mathbb{R}^{R_v \times M}$ is the projected anchor matrix for simplicity, and δ is the regularization parameter. Following [18], [22], $\mathbf{z}_i^T\mathbf{1} = 1$ and $z_{ij} \geq 0$ are the normalization and nonnegative constraints for the sparsity control of $\mathbf{Z}$. $\mathbf{S}_t^v = \mathbf{X}^v\mathbf{X}^{v\mathrm{T}}$ represents the total scatter matrix and $\mathbf{W}^{vT}\mathbf{S}_t^v\mathbf{W}^v = \mathbf{I}$ denotes the constraint to make the projected data statistically uncorrelated. To adaptively measure the modality contributions, the modality weights $\{1/\alpha_v\}_{v=1}^V$ are added and the sum-to-one constraint and nonnegative constraint are imposed on $\boldsymbol{\alpha} = [\alpha_1, \ldots, \alpha_V]^{\mathrm{T}}$.

3) Optimal Compressed Anchor Graph Learning: When the number of samples is huge, conventional anchor-graph-based clustering methods also require certain time cost to perform partition on the $N \times M$ anchor graph. To further accelerate partition, we construct a compressed consensus anchor graph $\mathbf{S} \in \mathbb{R}^{M \times M}$ from projected anchors in different modalities. Since $M \ll N$, $\mathbf{S}$ is much smaller than $\mathbf{Z}$. Thus, it is much more efficient to perform partition on $\mathbf{S}$. According to multimodal projected anchors $\{\mathbf{U}^v\}_{v=1}^V$, we construct $\mathbf{S}$ from different modalities to measure their shared anchor-to-anchor neighboring relationships in an autoweighted manner. Besides, to perform direct clustering, we desire that $\mathbf{S}$ has exactly C connected components corresponding to C clusters. It can be achieved by imposing the rank constraint $\mathrm{rank}(\mathbf{L}_S) = M - C$ on the Laplacian matrix $\mathbf{L}_S = \mathbf{D}_S - (\mathbf{S}^{\mathrm{T}} + \mathbf{S})/2$ related to $\mathbf{S}$ and the degree matrix $\mathbf{D}_S$ with diagonal elements $D_{ii} = \sum_j (s_{ij} + s_{ji})/2$. With these considerations, optimal compressed anchor graph learning is formulated by

$$\begin{aligned}\min_{\mathbf{U}^v, \mathbf{S}, \beta_v} & \sum_{v=1}^{V} \frac{1}{\beta_v} \sum_{i,j=1}^{M} \left\|\mathbf{u}_i^v - \mathbf{u}_j^v\right\|_2^2 s_{ij} + \gamma\|\mathbf{S}\|_F^2 \\ \text{s.t.} \quad & \mathbf{s}_i^{\mathrm{T}}\mathbf{1} = 1, \quad s_{ij} \geq 0,\ \boldsymbol{\beta}^{\mathrm{T}}\mathbf{1} = 1,\ \beta_v > 0 \\ & \mathrm{rank}(\mathbf{L}_S) = M - C \end{aligned} \tag{5}$$

where γ is the regularization parameter. $\mathbf{s}_i^T\mathbf{1} = 1$ and $s_{ij} \geq 0$ are the normalization and nonnegative constraints to control the sparsity of $\mathbf{S}$ as in [18] and [22]. To quantify the modality contributions, the modality weights $\{1/\beta_v\}_{v=1}^V$ are adaptively updated, and the sum-to-one constraint and nonnegative constraint are imposed on $\boldsymbol{\beta} = [\beta_1, \ldots, \beta_V]^{\mathrm{T}}$.

4) Overall Objective Function: The separated learning of the original and compressed consensus anchor graphs may result in a suboptimal result. Thus, we combine (4) and (5) into a unified framework to formulate the overall objective function of the proposed DGLAP as follows:

$$\min_{\substack{\mathbf{W}^v, \mathbf{U}^v, \mathbf{Z}, \\ \mathbf{S}, \alpha_v, \beta_v}} \sum_{v=1}^{V} \frac{1}{\alpha_v} \sum_{i=1}^{N}\sum_{j=1}^{M} \left\|\mathbf{W}^{vT}\mathbf{x}_i^v - \mathbf{u}_j^v\right\|_2^2 z_{ij} + \delta\|\mathbf{Z}\|_F^2$$

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$$+\sum_{v=1}^{V}\frac{1}{\beta_v}\sum_{i,j=1}^{M}\left\|\mathbf{u}_i^v-\mathbf{u}_j^v\right\|_2^2 s_{ij}+\gamma\|\mathbf{S}\|_F^2$$
$$\text{s.t.}\quad \mathbf{z}_i^{\mathrm{T}}\mathbf{1}=1,\quad z_{ij}\geq 0,\ \mathbf{s}_i^{\mathrm{T}}\mathbf{1}=1,\ s_{ij}\geq 0,\ \boldsymbol{\alpha}^{\mathrm{T}}\mathbf{1}=1$$
$$\alpha_v>0$$
$$\boldsymbol{\beta}^{\mathrm{T}}\mathbf{1}=1,\quad \beta_v>0,\ \mathbf{W}^{vT}\mathbf{S}_t^v\mathbf{W}^v=\mathbf{I}$$
$$\operatorname{rank}(\mathbf{L}_S)=M-C. \tag{6}$$

It should be noted that the constraint $\operatorname{rank}(\mathbf{L}_S)=M-C$ is nonlinear and difficult to solve. To address this issue, this constraint can be relaxed with the ith smallest eigenvalue of $\mathbf{L}_S$ denoted by $\sigma_i(\mathbf{L}_S)$. Based on the positive semi-definite property of $\mathbf{L}_S$, we can get $\sigma_i(\mathbf{L}_S)\geq 0$ and the rank constraint can be replaced by $\sum_{i=1}^{C}\sigma_i(\mathbf{L}_S)=0$. According to Ky Fan's theorem [38], we have $\sum_{i=1}^{C}\sigma_i(\mathbf{L}_S)=\min_{\mathbf{F}\in\mathbb{R}^{M\times C},\ \mathbf{F}^{\mathrm{T}}\mathbf{F}=\mathbf{I}}\operatorname{Tr}(\mathbf{F}^{\mathrm{T}}\mathbf{L}_S\mathbf{F})$. Thus, the overall objective function of the proposed DGLAP is finally formulated by

$$\min_{\substack{\mathbf{W}^v,\mathbf{U}^v,\mathbf{Z},\\ \mathbf{S},\mathbf{F},\alpha_v,\beta_v}}\sum_{v=1}^{V}\frac{1}{\alpha_v}\sum_{i=1}^{N}\sum_{j=1}^{M}\left\|\mathbf{W}^{vT}\mathbf{x}_i^v-\mathbf{u}_j^v\right\|_2^2 z_{ij}+\delta\|\mathbf{Z}\|_F^2$$
$$+\sum_{v=1}^{V}\frac{1}{\beta_v}\sum_{i,j=1}^{M}\left\|\mathbf{u}_i^v-\mathbf{u}_j^v\right\|_2^2 s_{ij}+\gamma\|\mathbf{S}\|_F^2$$
$$+2\lambda\operatorname{Tr}\left(\mathbf{F}^{\mathrm{T}}\mathbf{L}_S\mathbf{F}\right)$$
$$\text{s.t.}\quad \mathbf{z}_i^{\mathrm{T}}\mathbf{1}=1,\ z_{ij}\geq 0,\ \mathbf{s}_i^{\mathrm{T}}\mathbf{1}=1,\ s_{ij}\geq 0,\ \boldsymbol{\alpha}^{\mathrm{T}}\mathbf{1}=1$$
$$\alpha_v>0$$
$$\boldsymbol{\beta}^{\mathrm{T}}\mathbf{1}=1,\ \beta_v>0,\ \mathbf{W}^{vT}\mathbf{S}_t^v\mathbf{W}^v=\mathbf{I},\ \mathbf{F}\in\mathbb{R}^{M\times C}$$
$$\mathbf{F}^{\mathrm{T}}\mathbf{F}=\mathbf{I}. \tag{7}$$

where the regularization parameter λ should be large enough to satisfy $\sum_{i=1}^{C}\sigma_i(\mathbf{L}_S)=0$ such that $\mathbf{S}$ will have exactly C connected components.

5) Clustering Inference Strategy With Affinity Propagation: Based on the exactly C connected components in $\mathbf{S}$, M multimodal anchors can be easily and exactly partitioned into C clusters. Furthermore, we use these anchors with clustering labels and the affinity values between pixels and theses anchors as recording in $\mathbf{Z}$ to obtain the clustering results for all the pixels. Specifically, given a multimodal pixel, we can determine its label to be consistent with the label of a structured anchor according to the maximum affinity between this pixel and the connected anchors. Thus, N multimodal pixels can be directly partitioned into C clusters based on clustering labels of anchors and affinity propagation from $\mathbf{Z}$. Fig. 3 displays the efficient strategy with affinity propagation to infer clustering labels for all the pixels. It is much more efficient to learn clustering results from representative anchors in $\mathbf{S}$ than direct requiring exactly connected components in $\mathbf{Z}$.

B. Optimization

Due to the coupling of all the variables in (7), it is difficult to provide a direct solution for joint optimization. To solve problem (7), an alternating optimization algorithm is designed by updating each variable while fixing the others.

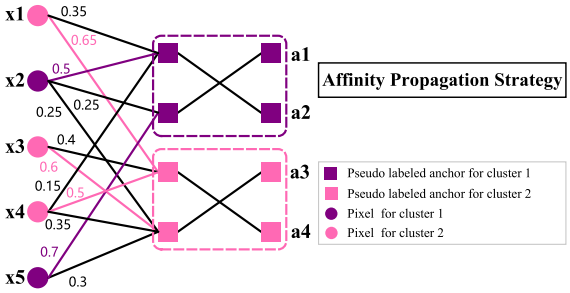


Fig. 3. Example of clustering inference strategy with affinity propagation. In the compressed anchor graph, four anchors are categorized into two clusters. Pixels are also divided into the two clusters based on the maximum affinity with the connected anchors in the anchor graph.

Algorithm 1 DGLAP

Input: Multimodal dataset $\{\hat{\mathcal{X}}^v\}_{v=1}^V$, cluster size C, superpixel size M, nearest denoising neighbor size L, projected dimension for each modality R_v;

Obtain $\{\mathbf{X}^v\}_{v=1}^V$ by performing spatial denoising;
Obtain $\{\mathbf{A}^v\}_{v=1}^V$ from spatial anchor generation;
Initialize $\mathbf{A}$, $\mathbf{Z}$; Initialize α_v and β_v with $\frac{1}{V}$;
repeat
  Update $\{\mathbf{W}^v\}_{v=1}^V$ by solving (12);
  Update $\{\alpha_v\}_{v=1}^V$ using (16);
  Update $\mathbf{Z}$ by solving (10);
  while not converge **do**
    Update $\mathbf{S}$ by solving (21);
    Update $\mathbf{F}$ by solving (19);
  end while
  Update $\{\beta_v\}_{v=1}^V$ using (18);
  Update $\{\mathbf{U}^v\}_{v=1}^V$ using (24);
until Convergence

Output: Determine clustering labels for pixels by a clustering inference strategy based on $\mathbf{Z}$ and $\mathbf{S}$.

1) Updating $\mathbf{Z}$*:* When $\mathbf{W}^v$, $\mathbf{U}^v$, $\mathbf{S}$, $\mathbf{F}$, α_v, and β_v are fixed, the optimization for $\mathbf{Z}$ becomes

$$\min_{\mathbf{Z}}\sum_{v=1}^{V}\frac{1}{\alpha_v}\sum_{i=1}^{N}\sum_{j=1}^{M}\left\|\mathbf{W}^{vT}\mathbf{x}_i^v-\mathbf{u}_j^v\right\|_2^2 z_{ij}+\delta\sum_{i=1}^{N}\sum_{j=1}^{M}z_{ij}^2$$
$$\text{s.t.}\quad \mathbf{z}_i^{\mathrm{T}}\mathbf{1}=1,\quad z_{ij}\geq 0. \tag{8}$$

Denoting $b_{ij}=\sum_{v=1}^{V}(1/\alpha_v)\|\mathbf{W}^{v\top}x_i^v-\mathbf{u}_j^v\|_2^2$, problem (8) can be solved by independently updating $\mathbf{z}_i$ as follows:

$$\min_{\mathbf{z}_i}\sum_{j=1}^{M}\left(b_{ij}z_{ij}+\delta z_{ij}^2\right),\quad \text{s.t.}\ \mathbf{z}_i^{\mathrm{T}}\mathbf{1}=1,\ z_{ij}\geq 0. \tag{9}$$

Then, problem (9) becomes

$$\min_{\mathbf{z}_i}\frac{1}{2}\left\|\mathbf{z}_i+\frac{\mathbf{b}_i}{2\delta}\right\|_2^2,\quad \text{s.t.}\ \mathbf{z}_i^{\mathrm{T}}\mathbf{1}=1,\ z_{ij}\geq 0. \tag{10}$$

As proved in [39], there is a closed-form solution to problem (10).

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2) Updating $\mathbf{W}^v$*:* When $\mathbf{Z}$, $\mathbf{U}^v$, $\mathbf{S}$, $\mathbf{F}$, β_v, and α_v are fixed, we can derive the optimization problem for $\mathbf{W}^v$ as follows:

$$\min_{\mathbf{W}^{v\mathrm{T}}\mathbf{S}_i^v\mathbf{W}^v=I} \sum_{i=1}^{N}\sum_{j=1}^{M}\left\|\mathbf{W}^{vT}\mathbf{x}_i^v-\mathbf{u}_j^v\right\|_2^2 z_{ij}. \tag{11}$$

According to [40] and $\mathbf{U}^v=\mathbf{W}^{v\mathrm{T}}\mathbf{A}^v$, (11) can be rewritten as

$$\min_{\mathbf{W}^{v\mathrm{T}}\mathbf{S}_i^v\mathbf{W}^v=I} \mathrm{Tr}\left(\mathbf{W}^{v\mathrm{T}}\mathbf{G}\mathbf{W}^v\right) \tag{12}$$

where $\mathbf{G}=\mathbf{X}^v\mathbf{D}\mathbf{X}^{v\mathrm{T}}-2\mathbf{X}^v\mathbf{Z}\mathbf{A}^{v\mathrm{T}}+\mathbf{A}^v\mathbf{D}_{\mathrm{col}}\mathbf{A}^{v\mathrm{T}}$, $\mathbf{D}=\mathrm{diag}(\sum_{j=1}^{M}z_{ij})$, and $\mathbf{D}_{\mathrm{col}}=\mathrm{diag}(\sum_{i=1}^{N}z_{ij})$. The optimal $\mathbf{W}^v$ is formed by the eigenvectors of $\mathbf{S}_i^{v-1}\mathbf{G}$ corresponding to R_v smallest eigenvalues.

3) Updating α_v*:* When $\mathbf{W}^v$, $\mathbf{U}^v$, $\mathbf{S}$, $\mathbf{F}$, β_v, and $\mathbf{Z}$ are fixed, the optimization for $\{\alpha_v\}_{v=1}^V$ is equal to

$$\begin{aligned}&\min_{\{\alpha_v\}_{v=1}^V}\sum_{v=1}^{V}\frac{1}{\alpha_v}\sum_{i=1}^{N}\sum_{j=1}^{M}\left\|\mathbf{W}^{vT}\mathbf{x}_i^v-\mathbf{u}_j^v\right\|_2^2 z_{ij}\\&\text{s.t.}\quad \boldsymbol{\alpha}^{\mathrm{T}}\mathbf{1}=1,\ \alpha_v>0.\end{aligned} \tag{13}$$

Denoting $h^v=(\sum_{i=1}^{N}\sum_{j=1}^{M}\|\mathbf{W}^{vT}\mathbf{x}_i^v-\mathbf{u}_j^v\|_2^2 z_{ij})^{1/2}$, problem (13) can be rewritten by

$$\min_{\{\alpha_v\}_{v=1}^V}\sum_{v=1}^{V}\frac{(h^v)^2}{\alpha_v},\qquad \text{s.t.}\quad \sum_{v=1}^{V}\alpha_v=1,\ \alpha_v>0. \tag{14}$$

According to Cauchy–Schwartz's inequality, we have

$$\sum_{v=1}^{V}\frac{(h^v)^2}{\alpha_v}=\left(\sum_{v=1}^{V}\frac{(h^v)^2}{\alpha_v}\right)\left(\sum_{v=1}^{V}\alpha_v\right)\geqslant\left(\sum_{v=1}^{V}h^v\right)^2. \tag{15}$$

The equality condition of (15) holds if and only if $((h^1/(\sqrt{\alpha_1})))/(\sqrt{\alpha_1})=((h^2/(\sqrt{\alpha_2})))/(\sqrt{\alpha_2})=\cdots=((h^V/(\sqrt{\alpha_V})))/(\sqrt{\alpha_V})$. Thus, we have $(h^1/\alpha_1)=(h^2/\alpha_2)=\cdots=(h^V/\alpha_V)=p$. Furthermore, we derive $\sum_{v=1}^{V}\alpha_v=\sum_{v=1}^{V}(h^v/p)=1$ and $p=\sum_{v=1}^{V}h^v$. Based on these conditions, we can update α_v by

$$\alpha_v=\frac{h^v}{\sum_{v=1}^{V}h^v}. \tag{16}$$

4) Updating β_v*:* When $\mathbf{W}^v$, $\mathbf{U}^v$, $\mathbf{F}$, $\mathbf{S}$, $\mathbf{Z}$, and α_v are fixed, the optimization for $\{\beta_v\}_{v=1}^V$ is equivalent to

$$\begin{aligned}&\min_{\{\beta_v\}_{v=1}^V}\sum_{v=1}^{V}\frac{1}{\beta_v}\sum_{i,j=1}^{M}\left\|\mathbf{u}_i^v-\mathbf{u}_j^v\right\|_2^2 s_{ij}\\&\text{s.t.}\quad \boldsymbol{\beta}^{\mathrm{T}}\mathbf{1}=1,\ \beta_v>0.\end{aligned} \tag{17}$$

Similar to the solution of α_v, we can update β_v by

$$\beta_v=\frac{e^v}{\sum_{v=1}^{V}e^v} \tag{18}$$

where $e^v=(\sum_{i,j=1}^{M}\|\mathbf{u}_i^v-\mathbf{u}_j^v\|_2^2 s_{ij})^{1/2}$.

5) Updating $\mathbf{F}$*:* When $\mathbf{W}^v$, $\mathbf{U}^v$, $\mathbf{S}$, $\mathbf{Z}$, β_v, and α_v are fixed, the optimization of $\mathbf{F}$ becomes

$$\min_{\mathbf{F}\in\mathbb{R}^{M\times C},\,\mathbf{F}^{\mathrm{T}}\mathbf{F}=I}\mathrm{Tr}\left(\mathbf{F}^{\mathrm{T}}\mathbf{L}_S\mathbf{F}\right). \tag{19}$$

The optimal solution of $\mathbf{F}$ is composed of the eigenvectors of $\mathbf{L}_S$ corresponding to C smallest eigenvalues.

6) Updating $\mathbf{S}$*:* When $\mathbf{W}^v$, $\mathbf{U}^v$, $\mathbf{F}$, $\mathbf{Z}$, α_v, and β_v are fixed, based on $\sum_{i,j=1}^{M}\|\mathbf{f}_i-\mathbf{f}_j\|_2^2 s_{ij}=2\,\mathrm{Tr}(\mathbf{F}^{\mathrm{T}}\mathbf{L}_S\mathbf{F})$, the optimization for $\mathbf{S}$ can be transformed into

$$\begin{aligned}\min_{\mathbf{S}}&\sum_{v=1}^{V}\frac{1}{\beta_v}\sum_{i,j=1}^{M}\left\|\mathbf{u}_i^v-\mathbf{u}_j^v\right\|_2^2 s_{ij}\\&+\gamma\sum_{i,j=1}^{M}s_{ij}^2+\lambda\sum_{i,j=1}^{M}\left\|\mathbf{f}_i-\mathbf{f}_j\right\|_2^2 s_{ij}\\&\text{s.t.}\quad \mathbf{s}_i^{\mathrm{T}}\mathbf{1}=1,\quad s_{ij}\geq 0.\end{aligned} \tag{20}$$

Denoting $c_{ij}=\sum_{v=1}^{V}(1/\beta_v)\|\mathbf{u}_i^v-\mathbf{u}_j^v\|_2^2+\lambda\|\mathbf{f}_i-\mathbf{f}_j\|_2^2$, problem (20) can be solved by independently updating $\mathbf{s}_i$ as follows:

$$\min_{\mathbf{s}_i}\frac{1}{2}\left\|\mathbf{s}_i+\frac{\mathbf{c}_i}{2\gamma}\right\|_2^2,\qquad \text{s.t.}\quad \mathbf{s}_i^{\mathrm{T}}\mathbf{1}=1,\ s_{ij}\geq 0. \tag{21}$$

As proved in [41], (21) can be solved with a closed-form solution.

7) Updating $\mathbf{U}^v$*:* When $\mathbf{W}^v$, $\mathbf{F}$, $\mathbf{S}$, $\mathbf{Z}$, α_v, and β_v are fixed, the optimization of $\mathbf{U}^v$ is equivalent to

$$\min_{\mathbf{U}^v}\frac{1}{\alpha_v}\sum_{i=1}^{N}\sum_{j=1}^{M}\left\|\mathbf{W}^{v\mathrm{T}}\mathbf{x}_i^v-\mathbf{u}_j^v\right\|_2^2 z_{ij}+\frac{1}{\beta_v}\sum_{i,j=1}^{M}\left\|\mathbf{u}_i^v-\mathbf{u}_j^v\right\|_2^2 s_{ij}. \tag{22}$$

Using the previously derived equations and removing irrelevant terms, we can derive the following equation:

$$\begin{aligned}\min_{\mathbf{U}^v}\mathrm{Tr}\Bigg(&-\frac{2}{\alpha_v}\mathbf{W}^{v\mathrm{T}}\mathbf{X}^v\mathbf{Z}\mathbf{U}^{v\mathrm{T}}\\&+\frac{1}{\alpha_v}\mathbf{U}^v\mathbf{D}_{\mathrm{col}}\mathbf{U}^{v\mathrm{T}}+\frac{2}{\beta_v}\mathbf{U}^v\mathbf{L}_S\mathbf{U}^{v\mathrm{T}}\Bigg).\end{aligned} \tag{23}$$

The optimal solution of $\mathbf{U}^v$ can be obtained by taking the derivative and setting it to zero. Thus, we have

$$\mathbf{U}^v=\beta_v\mathbf{W}^{v\mathrm{T}}\mathbf{X}^v\mathbf{Z}^v(\beta_v\mathbf{D}_{\mathrm{col}}+2\alpha_v\mathbf{L}_S)^{-1}. \tag{24}$$

In summary, the entire optimization process is listed in Algorithm 1.

C. Complexity Analysis

1) Time Complexity: The time complexity of the proposed method with Algorithm 1 is attributed to updating each variable. To solve $\{\mathbf{W}^v\}_{v=1}^V$, it takes $\mathcal{O}(\sum_{v=1}^{V}B_v^3)$ to achieve eigenvalue decomposition and matrix inversion. To learn $\mathbf{Z}$, it needs $\mathcal{O}(NMk)$ for calculation. The update of $\mathbf{S}$ requires $\mathcal{O}(M^2k)$ to get the closed-form solution. To solve $\mathbf{F}$, it requires $\mathcal{O}(M^2C)$ to calculate the eigenvectors of $\mathbf{L}_S$. To update $\{\mathbf{U}^v\}_{v=1}^V$, it costs $\mathcal{O}(RBN+RNM+M^3+RM^2)$, where $B=\sum_{v=1}^{V}B_v$ and $R=\sum_{v=1}^{V}R_v$. As for α_v and β_v, their update trivially requires $\mathcal{O}(1)$. For clustering inference strategy, it takes $\mathcal{O}(NM)$ to get the clustering results. Since the proposed method involves a subalternating system, we set the number of iterations for the inner and outer loops as T_1 and T_2, respectively. Considering that $R<B\approx M$, $M\ll N$, the overall time complexity is $\mathcal{O}(((M^2k+M^2C)T_1+NMR)T_2)$. Denoting $T=T_1T_2$, the time complexity can be simplified as $\mathcal{O}(NMBT)$. Thus, the proposed method shows a nearly linear time complexity of $\mathcal{O}(N)$.

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TABLE I
SUMMARY OF MULTIMODAL REMOTE SENSING DATASETS

Datasets	Sizes	Clusters	Modality Details		
			Modality1	Modality2	Modality3
Houston	349×1905	15	HS(144)	MS(8)	–
Berlin	1723×476	8	HS(244)	SAR(4)	–
Augsburg	332×485	7	HS(180)	SAR(4)	DSM(1)

2) Space Complexity: The major space consumption of the proposed method comes from the matrices $\mathbf{W}^v \in \mathbb{R}^{B_v \times R_v}$, $\mathbf{Z} \in \mathbb{R}^{N \times M}$, $\mathbf{S} \in \mathbb{R}^{M \times M}$, $\mathbf{F} \in \mathbb{R}^{M \times C}$. and $\mathbf{U}^v \in \mathbb{R}^{R_v \times M}$. Thus, the major space complexity is $\mathcal{O}(NM + M(M + C + R) + BR)$. This is also almost linear to the number of samples $\mathcal{O}(N)$.

IV. EXPERIMENTS

A. Experimental Details

1) Datasets: Three datasets are adopted for evaluation, including Houston, Berlin, and Augsburg. The Houston scene with 349 × 1905 pixels was gathered from an HS image with 144 spectral bands and an MS image with eight channels. There are 15 categories of land covers recorded in this scene. The Berlin scene was recorded from an HS image with 244 spectral bands by HyMap sensor and an SAR image with four channels by Sentinel-1 product. There are eight classes of land covers recorded within 1723 × 476 pixels. The Augsburg scene with three different sources was collected from an HS image with 180 spectral bands by HySpex sensor, an SAR image with four features by Sentinel-1 satellite, and a DSM image with one band by DLR-3K system. There are 332 × 485 pixels containing seven types of land covers. Detailed information of these datasets is summarized in Table I.

2) Compared Methods: As for as we know, there are only two pixelwise clustering methods designed for multimodal remote sensing data. To validate the superiority of the proposed method, we adopt five single-modal clustering methods designed for single-modal remote sensing data, two multimodal clustering methods designed for multimodal remote sensing data, and two multimodal clustering methods designed for generic multimodal data as compared methods. For single-modal remote sensing clustering methods, they are fast spectral clustering with anchor graph (FSCAG) [18], scalable graph-based clustering with nonnegative relaxation (SGCNR) [22], superpixel global and local similarity graph-based clustering (SGLSC) [35], superpixel neighborhood contrastive subspace clustering (NCSC) network [42], and multiscale spectral-spatial association network (MS^{2}A-Net) [43]. For multimodal remote sensing clustering methods, they are multisensor deep clustering (MDC) [28] and transformer-based multimodal prototypical contrastive clustering (TMPCC) [29]. As for multimodal generic clustering methods, they are adversarial multiview clustering (AMvC) network [44] and multilevel feature learning for contrastive multiview clustering (MFLVC) [45]. Details of these methods are demonstrated in Table II.

3) Setup: The registration issue for multimodal images is to learn the aligned images at the same spatial coordinates. Different from the multimodal images used for registration, the multimodal images for clustering analysis in our experiments are with the same size in different modalities and have been aligned to ensure that the pixels at the same spatial position share the same ground-truth labels. Thus, there is no need to perform registration in our experiments. For fair comparison, images in different modalities are all used as input in the experiment, and modality exploration manners adopted in different methods are shown in Table II. Specifically, images in different modalities are stacked along the third dimension and the concatenated image is fed as input for single-modal methods, while images in different modalities are directly used as input for multimodal methods. In our settings, hyperparameters are needed to adjust. When the number of connected components in $\mathbf{S}$ is greater or smaller than C, we empirically perform $\lambda = \lambda/2$ or $\lambda = 2 * \lambda$ for parameter λ. To achieve projection learning, the projected dimension R_v is tuned from $[B_v/4{:}5{:}B_v/2]$ for high-dimensional modalities and varies from $[1{:}2{:}B_v]$ for low-dimensional modalities. The number of superpixels M is tuned from [150:5:300], and the number of denoising neighbors L varies from [3:1:15]. The source codes of the compared methods are downloaded from the websites or requested from the authors. Based on the original papers, the parameters of the compared methods are set to achieve the best performance. To quantify the clustering performance, eight common metrics are used for evaluation, including clustering accuracy (ACC), normalized mutual information (NMI), purity, adjusted rand index (ARI), Kappa coefficient, F-score, Precision, and Recall. All the reported results are averaged from ten repeated trails on a server with two Intel[1] Xeon[1] Silver 4316 CPUs 2.30 GHz, 256 G RAM, NVIDIA A100 80-GB GPU.

B. Comparison Results

1) Clustering Performance: Table III reports the clustering performance of all the compared methods on three datasets. The proposed DGLAP achieves the best results on these datasets as highlighted in boldface. Compared with single-modal methods, DGLAP significantly outperforms SGLSC for the best shallow method and MS^{2}A-Net for the best deep method with 25.89% and 20.86% improvements in terms of ACC on the Berlin dataset. This indicates that DGLAP with fusion strategy is effective for fusing multimodal data compared with simple feature concatenation. Besides, DGLAP is remarkably superior to MFLVC for the best multimodal generic method and MDC for the best multimodal remote sensing method with 18.20% and 17.16% improvements in terms of ACC on the Houston dataset. This verifies that DGLAP is good at spatial information exploitation with denoising as well as anchor selection and geometric structure exploration to uncover the clustering structure. Since MDC is only suitable for multimodal data with two modalities, the results are unavailable on the Augsburg dataset with three modalities that are denoted as "N/A." As shown in Table III, other results on multimodal remote sensing image datasets also validate the superiority of DGLAP.

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TABLE II
DETAILED INFORMATION OF DIFFERENT COMPARED METHODS

Comparedmethods	Method application		Method type		Method design		Modality exploration		Information utilization			Source	
	Remote sensing	Generic data	Shallow learning	Deep learning	Single modality	Multiple modality	Concatenation	Fusion	Spectrum	Spatiality	Geometry	Venue	Year
FSCAG [18]	✓		✓		✓		✓		✓	✓	✓	GRSL	2017
SGCNR [22]	✓		✓		✓		✓		✓		✓	TGRS	2019
SGLSC [35]	✓		✓		✓		✓		✓	✓	✓	TGRS	2022
NCSC [42]	✓			✓	✓		✓		✓	✓		TGRS	2022
MS^{2}A-Net [43]	✓			✓	✓		✓		✓	✓		JSTARS	2022
MDC [28]	✓			✓		✓		✓	✓	✓		JSTARS	2022
TMPCC [29]	✓			✓		✓		✓	✓	✓		Inf. Sci.	2023
AMvC [44]		✓		✓		✓		✓	✓			TNNLS	2022
MFLVC [45]		✓		✓		✓		✓	✓			CVPR	2022
DGLAP (Ours)	✓		✓			✓		✓	✓	✓	✓	–	–

TABLE III
CLUSTERING PERFORMANCE OF DIFFERENT METHODS ON MULTIMODAL REMOTE SENSING DATASETS. THE OPTIMAL RESULTS ARE SHOWN IN BOLD

Datasets	Metrics	Single-modal Methods					Multimodal Methods				
		FSCAG	SGCNR	SGLSC	NCSC	MS^{2}A-Net	MDC	TMPCC	AMvC	MFLVC	DGLAP
Houston	ACC	0.4714	0.4416	0.3616	0.3085	0.3879	0.4310	0.4211	0.3601	0.4256	**0.5203**
	NMI	0.4978	0.4726	0.3890	0.4012	0.4115	0.4791	0.4411	0.3854	0.5156	**0.5510**
	Purity	0.4970	0.4733	0.3892	0.3480	0.4106	0.4665	0.4211	0.3714	0.4648	**0.5344**
	ARI	0.3058	0.2643	0.1909	0.1757	0.2464	0.2781	0.2626	0.1857	0.2908	**0.3192**
	Kappa	0.4298	0.3975	0.3104	0.2576	0.3383	0.3869	0.3752	0.3061	0.3847	**0.4810**
	F-score	0.3914	0.3746	0.3338	0.2681	0.3005	0.3699	0.3160	0.2983	0.3827	**0.4949**
	Precision	0.4283	0.4324	0.3594	0.2993	0.3595	0.4704	0.2946	0.3857	0.4446	**0.5721**
	Recall	**0.4385**	0.4129	0.3599	0.3209	0.3369	0.4149	0.4031	0.3354	0.4225	0.4291
Berlin	ACC	0.5079	0.4892	0.5348	0.2800	0.5711	0.5083	0.4455	0.5110	0.4165	**0.7216**
	NMI	0.3134	0.2785	0.3368	0.2491	0.3193	0.3187	0.4163	0.1850	0.3119	**0.4242**
	Purity	0.7199	0.6573	0.7252	0.6775	0.7117	0.6617	**0.7702**	0.6403	0.6836	0.7619
	ARI	0.2467	0.2251	0.3204	0.0672	0.3335	0.2621	0.2526	0.2160	0.2142	**0.4468**
	Kappa	0.3331	0.3135	0.3580	0.1617	0.3877	0.3034	0.3430	0.2188	0.2641	**0.4884**
	F-score	**0.3874**	0.3504	0.3491	0.2033	0.3719	0.2525	0.3748	0.2508	0.3327	0.3653
	Precision	0.4759	0.4173	0.4243	0.2569	0.3662	0.3467	0.4065	0.3023	0.3190	**0.4947**
	Recall	0.4287	0.4081	0.3872	0.2886	0.4104	0.3016	**0.5112**	0.2607	0.4423	0.3683
Augsburg	ACC	0.4463	0.3622	0.5399	0.3263	0.5702	N/A	0.6269	0.5008	0.5468	**0.8497**
	NMI	0.2895	0.2267	0.4477	0.1751	0.3252	N/A	0.4600	0.2720	0.3884	**0.6543**
	Purity	0.6088	0.5750	0.7903	0.5358	0.6518	N/A	0.7559	0.6046	0.7468	**0.8644**
	ARI	0.2231	0.1387	0.3962	0.0867	0.3545	N/A	0.4029	0.2317	0.3191	**0.6869**
	Kappa	0.2879	0.2138	0.4334	0.2161	0.4409	N/A	0.4996	0.3211	0.4390	**0.7772**
	F-score	0.3311	0.2955	0.3801	0.2369	0.3392	N/A	0.3264	0.2885	0.3693	**0.4777**
	Precision	0.4597	0.3656	0.4564	0.3357	0.3694	N/A	0.3434	0.3090	0.4203	**0.6562**
	Recall	0.3610	0.3835	0.4577	0.3297	0.3922	N/A	0.3466	0.3103	0.4109	**0.4596**

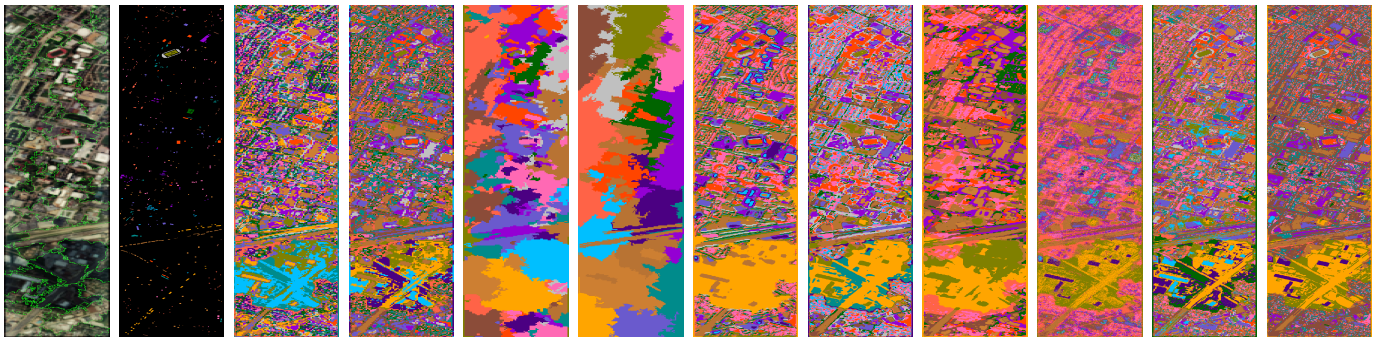
For fair comparison, images in different modalities are all utilized as input for the single-modal methods and multimodal methods. "N/A" means the results are unavailable on the Augsburg dataset since MDC is only suitable for multimodal data with two modalities.

TABLE IV
p-VALUES OF SIGNIFICANT DIFFERENCE BETWEEN DGLAP AND OTHER METHODS ON MULTIMODAL REMOTE SENSING DATASETS

Comparisons	Houston	Berlin	Augsburg
FSCAG-DGLAP	1.927E-06	8.668E-08	2.036E-24
SGCNR-DGLAP	1.822E-08	8.813E-09	1.891E-19
SGLSC-DGLAP	1.140E-270	6.852E-270	4.621E-274
NCSC-DGLAP	1.365E-13	3.508E-18	1.731E-22
MS^{2}A-Net-DGLAP	1.437E-13	7.294E-11	2.764E-14
MDC-DGLAP	8.039E-17	9.733E-15	N/A
TMPCC-DGLAP	1.035E-10	5.516E-19	2.301E-11
AMVC-DGLAP	1.670E-10	4.4843E-10	3.793E-19
MFLVC-DGLAP	5.624E-09	7.668E-06	7.662E-18

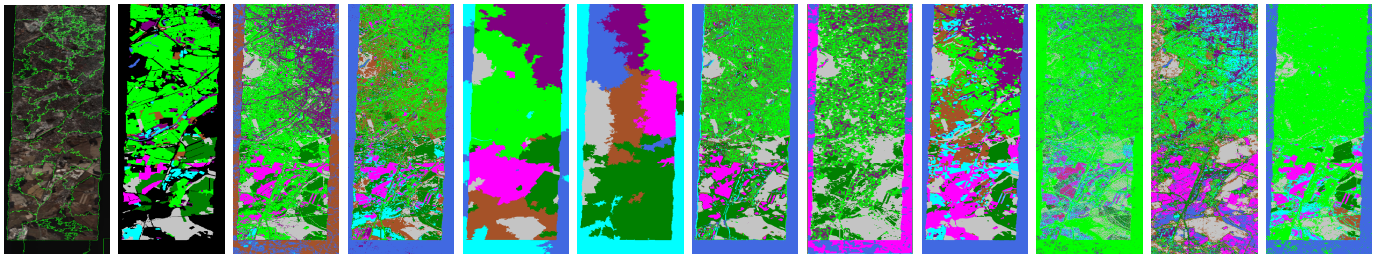
To verify whether DGLAP is significantly outperform other compared methods, we adopt a two-tailed t-test with 95% confidence to calculate the pairwise results in a statistical perspective. The two-tailed t-test is a statistical method used to determine whether there is a significant difference between the means of two groups of data. To perform the two-tailed t-test, two groups of data with the same size are independent and come from normal distribution. The experimental results used for the two-tailed t-test satisfy these prerequisites. The null hypothesis is that there is no difference between the two means, while the alternative hypothesis is that there is a difference between the two means. If the p-value of a pairwise comparison is less than 0.05, the null hypothesis is rejected in favor of the alternative. Table IV displays the p-values of pairwise comparisons to show the statistical results in terms of ACC. Since all the p-values of pairwise comparisons are much smaller than 0.05, we conclude that DGLAP consistently

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(a) (b) (c) (d) (e) (f) (g) (h) (i) (j) (k) (l)

Fig. 4. Clustering maps of the Houston dataset obtained by different methods. (a) False color image. (b) Ground truth. (c) FSCAG. (d) SGCNR. (e) SGLSC. (f) NCSC. (g) MS^{2}A-Net. (h) MDC. (i) TMPCC. (j) AMvC. (k) MFLVC. (l) DGLAP.



(a) (b) (c) (d) (e) (f) (g) (h) (i) (j) (k) (l)

Fig. 5. Clustering maps of the Berlin dataset obtained by different methods. (a) False color image. (b) Ground truth. (c) FSCAG. (d) SGCNR. (e) SGLSC. (f) NCSC. (g) MS^{2}A-Net. (h) MDC. (i) TMPCC. (j) AMvC. (k) MFLVC. (l) DGLAP.

demonstrates promising clustering performance with significant difference compared with other methods.

To visualize the clustering effect, Figs. 4–6 display the clustering maps of different methods on the three datasets. As superpixel-level methods, SGLSC and NCSC cannot describe more details about the spatial distribution of land covers in their clustering maps. As shown in Fig. 5, for the clustering map of the Berlin dataset obtained by the proposed method, there are some speckles in the black area of ground truth. The pixels in these speckles occupy a very small portion of all the pixels. As for the clustering maps of MS^{2}A-Net and TMPCC, there are many pixels misclassified and mixed in the light green area of ground truth. Thus, the proposed method achieves higher clustering accuracy and outperforms MS^{2}A-Net and TMPCC. Among pixel-level methods, the clustering maps of DGLAP are closest to the ground-truth maps. This is consistent with the high clustering results in Table III.

2) Running Time: Table V shows the running time consumed by different methods on the three datasets under the same environment. Among all the compared methods, FSCAG is the fastest because of its noniterative property. Except for FSCAG, DGLAP shows high efficiency with the least running time on the three datasets, even on the Berlin dataset with nearly one million pixels. Although SGLSC and NCSC rescale clustering at superpixel level, they are still slightly time-consuming. Due to GPU acceleration, MS^{2}A-Net and MFLVC are efficient but require more running time than DGLAP. It is obvious that TMPCC and AMvC are the most time-consuming methods because of their complex network training requiring much more learning iterations. Overall, DGLAP is superior in clustering performance and computational efficiency even for multimodal data with large-scale pixels.

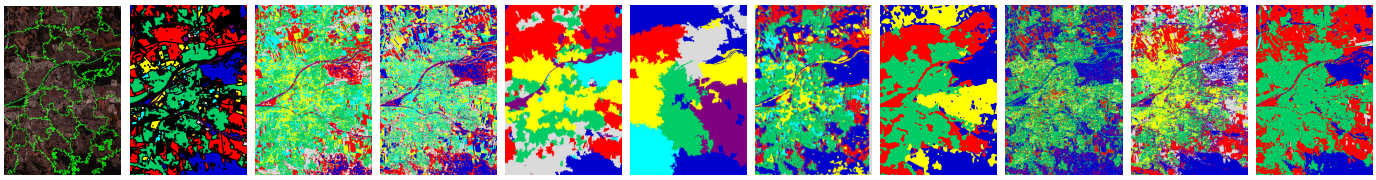
TABLE V

TIME COMPLEXITY AND RUNNING TIME (t/s) COMPARISONS OF DIFFERENT METHODS ON MULTIMODAL REMOTE SENSING DATASETS. THE SHORTEST RUNNING TIME IS SHOWN IN BOLD

Methods	Time Complexity	Runtime		
		Houston	Berlin	Augsburg
FSCAG	$\mathcal{O}(NMB)$	**18.55**	**22.45**	**14.17**
SGCNR	$\mathcal{O}(NMB)$	177.35	145.34	23.71
SGLSC	$\mathcal{O}(M^2BT)$	1300.2	986.15	222.79
NCSC	$\mathcal{O}(M^2BT)$	259.78	278.3	61.34
MS^{2}A-Net	$\mathcal{O}(NWT)$	170.28	346.5	45.1
MDC	$\mathcal{O}(NWT)$	230.04	433.52	N/A
TMPCC	$\mathcal{O}(NWT)$	18683.05	21197.17	6352.55
AMvC	$\mathcal{O}(NWT)$	17147.98	17609.16	4038.48
MFLVC	$\mathcal{O}(NWT)$	162.01	312.07	168.9
DGLAP	$\mathcal{O}(NMBT)$	41.93	61.43	21.72

To analyze the efficiency of the compared methods, Table V also displays their time complexities. Here, N, B, M, W, and T represent the number of pixels, bands, superpixels or anchors, network weights, and iterations, respectively. Table V shows that the compared methods exhibit linear growth in time complexity with respect to the number of pixels except for SGLSC and NCSC. The time complexity of SGLSC and NCSC increases quadratically with the number of superpixels, meaning that the computational cost significantly increases as the number of superpixels increases. Since DGLAP exhibits linear time complexity with respect to the number of pixels, it shows highly efficient in processing large datasets.

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(a) (b) (c) (d) (e) (f) (g) (h) (i) (j) (k)

Fig. 6. Clustering maps of the Augsburg dataset obtained by different methods. (a) False color image. (b) Ground truth. (c) FSCAG. (d) SGCNR. (e) SGLSC. (f) NCSC. (g) MS^{2}A-Net. (h) TMPCC. (i) AMvC. (j) MFLVC. (k) DGLAP.

TABLE VI
ABLATION STUDIES ON MULTIMODAL REMOTE SENSING DATASETS

Settings	Components					Augsburg			Berlin			Huston		
	SG	SD	SAS	PL	AMW	ACC	NMI	Purity	ACC	NMI	Purity	ACC	NMI	Purity
S1	✗	✗	✗	✗	✗	0.4207	0.2393	0.4859	0.3607	0.1681	0.4937	0.3480	0.4206	0.3611
S2	✓	✓	✗	✗	✗	0.5529	0.3045	0.5713	0.4875	0.1118	0.5980	0.3824	0.4335	0.4109
S3	✓	✗	✓	✗	✗	0.5212	0.2509	0.5382	0.5622	0.1660	0.6066	0.4228	0.4601	0.4517
S4	✓	✓	✓	✗	✗	0.5963	0.3274	0.6029	0.5936	0.1941	0.6214	0.4521	0.5025	0.4720
S5	✓	✓	✓	✓	✗	0.7254	0.5150	0.7880	0.6310	0.2121	0.6472	0.4769	0.5491	0.4763
DGLAP	✓	✓	✓	✓	✓	**0.8497**	**0.6543**	**0.8644**	**0.7216**	**0.4242**	**0.7619**	**0.5203**	**0.5651**	**0.5344**

SG: Superpixel generation. SD: Spatial denoising. SAS: Spatial anchor selection. PL: Projection learning. AMW: Adaptive modality weight.

TABLE VII
CLUSTERING PERFORMANCE WITH DIFFERENT DENOISING AND ANCHOR SELECTION STRATEGIES ON MULTIMODAL REMOTE SENSING DATASETS

Datasets	Metrics	Denoising		Anchor selection	
		MF	Our SD	K-means	Our SAS
Houston	ACC	0.4417	**0.5203**	0.4159	**0.5203**
	NMI	0.5255	**0.5651**	0.4991	**0.5651**
	Purity	0.4936	**0.5344**	0.4215	**0.5344**
Augsburg	ACC	0.7360	**0.8497**	0.5807	**0.8497**
	NMI	0.4995	**0.6543**	0.3625	**0.6543**
	Purity	0.7549	**0.8644**	0.5961	**0.8644**
Berlin	ACC	0.6447	**0.7216**	0.6315	**0.7216**
	NMI	0.2585	**0.4242**	0.2744	**0.4242**
	Purity	0.6851	**0.7619**	0.6316	**0.7619**

MF: Mean filtering. SD: Spatial denoising. SAS: Spatial anchor selection.

TABLE VIII
CLUSTERING PERFORMANCE OF DIFFERENT GRAPH-BASED VARIANTS WITH RANK CONSTRAINT ON MULTIMODAL REMOTE SENSING DATASETS. THE BEST ACC AND SHORTEST RUNTIME ARE IN BOLD FONT

Datasets	Metrics	$rank(\mathbf{S})$	$rank(\mathbf{Z})$	DGLAP
Houston	ACC	0.4319	0.4774	**0.5203**
	Runtime (s)	**17.75**	625.96	41.93
Berlin	ACC	0.6588	**0.7262**	0.7216
	Runtime (s)	**32.01**	842.75	61.43
Augsburg	ACC	0.4501	0.5778	**0.8497**
	Runtime (s)	**5.17**	212.59	21.72

$rank(\mathbf{S})$, $rank(\mathbf{Z})$: Degraded variants with rank constraint on $\mathbf{S}$ and $\mathbf{Z}$ for superpixel-level clustering and pixel-level clustering.

C. Model Analyses

For a deep insight into the proposed DGLAP, we provide detailed model analyses at different perspectives.

1) Ablation Study: To investigate the effect of each component in DGLAP, we conduct ablation experiments with different settings by removing different components. Apart from the major dual graph learning, we use "✓" and "✗" to denote different settings containing a specific component or not. As shown in Table VI, setting "S1" shows the worst performance without considering any components. When performing both spatial denoising and anchor selection, setting "S4" is superior to setting "S2" with only spatial denoising and setting "S3" with only spatial anchor selection. By introducing projection learning, the performance of setting "S5" is improved than that of setting "S4." It is clear that DGLAP shows the best performance with at least 14.43%–7.64% improvement in terms of ACC and Purity than other settings by simultaneous considering all the valuable components to better exploit multimodal remote sensing data.

To further test the effect of our spatial denoising and anchor selection, we perform mean filtering for denoising [18] and K-means clustering for anchor selection [22] for comparison. As shown in Table VII, compared with mean filtering and K-means clustering, our spatial denoising and anchor selection are able to well use spatial information to achieve much higher clustering results. Besides, we also develop two degraded variants of DGLAP to verify the efficiency and efficacy of dual graph learning and clustering strategy with affinity propagation. They are "rank($\mathbf{S}$)" and "rank($\mathbf{Z}$)" to, respectively, learn the compressed anchor graph $\mathbf{S}$ and original anchor graph $\mathbf{Z}$ with rank constraint for superpixel-level clustering and pixel-level clustering. From Table VIII, "rank($\mathbf{S}$)" requires less execution time but shows very limited accuracy for superpixel-level clustering, while "rank($\mathbf{Z}$)" consumes much more running time and exhibits superior accuracy for pixel-level clustering. In contrast, DGLAP shows superior or comparable accuracy compared with "rank($\mathbf{Z}$)" at very low time cost, keeping a good balance between efficiency and efficacy.

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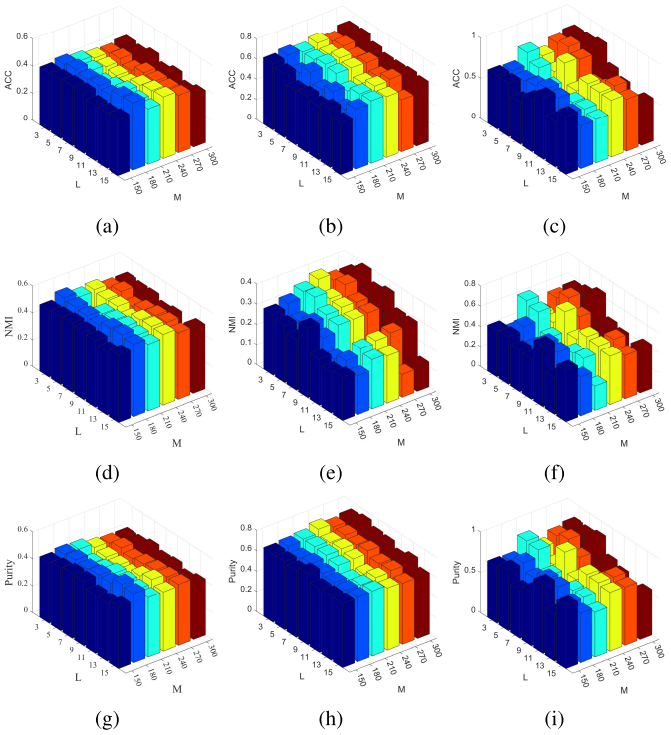


Fig. 7. Performance variance of DGLAP with different settings of M and L on multimodal remote sensing datasets. (a) Houston (ACC). (b) Berlin (ACC). (c) Augsburg (ACC). (d) Houston (NMI). (e) Berlin (NMI). (f) Augsburg (NMI). (g) Houston (Purity). (h) Berlin (Purity). (i) Augsburg (Purity).

2) Parameter Sensitivity Study: The numbers of superpixels and nearest denoising neighbors (M and L) are vital for DGLAP. To study their sensitivities, we tune M from [150:30:300] and L from [3:2:15]. As shown in Fig. 7, DGLAP obtains superior clustering results with different numbers of superpixels and nearest denoising neighbors in most cases. When assigning improper values for these two parameters, poor performance is incurred in a few cases. This can be explained that a suitable number of superpixels is able to well-explore spatial information from local regions, while an appropriate number of nearest denoising neighbors is beneficial to improve feature representation capability for further learning.

3) Modality Dimension and Contribution Study: It is important to learn the projected dimension and adaptive weight of each modality for DGLAP to fuse multimodal data. In Fig. 8, the X-axis records different modalities, while the left Y-axis describes the projected modality dimensions and the right Y-axis shows the adaptive modality weights to construct the two different graphs. The modality weights shown in Fig. 8 are adaptively learned in (7) to measure the contribution of each modality for constructing the original anchor graph and the compressed anchor graph. For the Augsburg dataset, DGLAP achieves the best performance by reducing the dimension from 180 to 70 for HS image and remaining the dimension for SAR and DSM images. The modality weights are largest for SAR image and much smaller for DSM and HS images to learn the original and compressed consensus anchor graphs. It is obvious that SAR image contributes more to constructing the dual graphs than DSM and HS images.

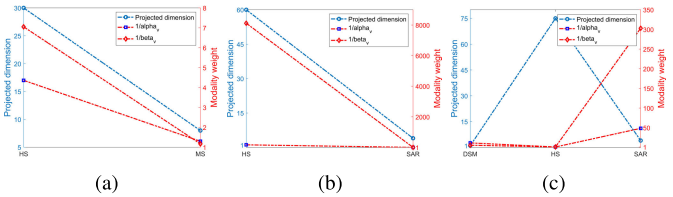


Fig. 8. Projected dimension and contribution of each modality on multimodal remote sensing datasets. (a) Houston. (b) Berlin. (c) Augsburg.

TABLE IX

CLUSTERING PERFORMANCE OF DGLAP WITH EACH SINGLE MODALITY AND ALL MODALITIES ON MULTIMODAL REMOTE SENSING DATASETS

Datasets	Metrics	DGLAP			
		Modality1	Modality2	Modality3	Fusion
Houston	ACC	0.4094	0.4058	N/A	**0.5203**
	NMI	0.4984	0.5059	N/A	**0.5510**
	Purity	0.4394	0.4640	N/A	**0.5344**
Berlin	ACC	0.6965	0.5662	N/A	**0.7216**
	NMI	0.3357	0.1136	N/A	**0.4242**
	Purity	0.7095	0.6126	N/A	**0.7619**
Augsburg	ACC	0.6683	0.4399	0.4446	**0.8497**
	NMI	0.5306	0.1563	0.1475	**0.6543**
	Purity	0.6861	0.4910	0.4805	**0.8644**

Fusion: All modalities are used for fusion learning and clustering.

TABLE X

FUSION EFFECT OF DGLAP WITH DIVERSE MODALITIES AS INPUT ON THE HOUSTON DATASET. THE BEST CLUSTERING RESULTS ARE IN BOLD FONT

Houston	Performance			Weight $\frac{1}{\alpha_v}$		Weight $\frac{1}{\beta_v}$	
	ACC	NMI	Purity	HS	HS/MS	HS	HS/MS
HS+HS	0.4092	0.4984	0.4392	2	2	2	2
HS+MS	**0.5203**	**0.5510**	**0.5344**	4.36	1.3	7.05	1.172

4) Modality Fusion Effect: To verify the modality fusion effect, DGLAP is applied to each modality individually without considering other modalities. In Table IX, "Modality1," "Modality2," and "Modality3" indicate that the first, second, and third modalities are used as input individually, while "Fusion" denotes that all the modalities are simultaneously used as input for fusion learning. Using only one modality for experiments, there is no doubt that the clustering performance is greatly reduced on the three datasets. When using all the modalities, DGLAP gains much higher clustering results. Compared with single modality with the highest results, DGLAP using all modalities gains 7.04%, 5.24%, and 17.83% improvements in terms of Purity on the three datasets, respectively. This can be further explained that the proposed fusion strategy in DGLAP is promising to fuse multimodal data for learning promotion. In summary, DGLAP shows its superiority of multimodal remote sensing image fusion and clustering.

To further demonstrate the adaptive weighting and fusion strategies, we have conducted experiments by replacing MS image with HS image on the Houston dataset. This means two same HS images of the Houston dataset are used as input for the proposed DGLAP. The corresponding results are shown

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in Table X. From Table X, given the Houston dataset with two same HS images as input for DGLAP, the clustering performance is almost the same as the case of only using HS image of Modality1 as input in Table IX. The modality weights of two same HS images are the same for constructing both the original and compressed anchor graphs, meaning that they have the same contribution. When conducting DGLAP on the Houston dataset with HS and MS images, the modality weights of HS image are larger than those of MS image for constructing the two graphs, meaning that HS image contributes more than MS image. However, this does not mean that MS image is insignificant, because a small weight for MS image can also provide important complementary information. Thus, both HS and MS images are crucial to multimodal fusion and learning. It further demonstrates the effectiveness and reasonability of the adaptive weighting and fusion strategies in DGLAP over the single-modality strategy.

V. Conclusion

In this article, we introduced a DGLAP method for multimodal remote sensing image clustering. Based on local region guidance, the original consensus anchor graph is constructed from the denoised pixels and generated anchors with projection learning and adaptive modality weighting. Meanwhile, the optimal compressed consensus anchor graph is constructed from the projected anchors with connectivity constraint and adaptive modality weighting. An alternating optimization strategy with linear complexity is devised for the proposed formulation to promote dual graph learning in a mutual reinforcement manner. According to the anchors distributed in the structured compressed anchor graph with exactly connected components, clustering results of pixels can be directly assigned to be same as the connected anchor with the maximum affinity as recording in the original anchor graph. Experiments have demonstrated the superiority of the proposed method.

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